Counting, Part 1

Session II · 14 September 2026 — thalidomide, rare events, and the HIV arithmetic

Class notes · an automated summary of the live session, structured for study. Where these notes and the assigned papers disagree, the papers win.

Key outcomes

Part one of the counting session. Two case studies: the thalidomide disaster as the landmark failure of pre-market safety testing, and the HIV epidemic as the live illustration of the core formula prevalence = incidence × duration — why the number of people living with HIV is at its historical maximum precisely because treatment works. The session also returned to rare adverse events (myocarditis after COVID-19 vaccination) to fix the idea that a study's size sets the rarest event it can see.

Thalidomide: the safety failure

- A sedative that reached the market in 1957, sold in 46 countries, often without prescription, and widely taken in pregnancy
- Caused a teratogenic effect — developmental harm to the embryo from a toxic exposure, distinct from genetic mutation
- The core failure: the drug was never tested for safety in pregnancy before release. Ethics kept pregnant women out of trials, but animal studies on pregnant subjects could have been run — and were not
- The catastrophe was not the absence of any regulatory option; it was the failure to measure the probability of harm in the specific subgroup
- A comprehensive historical analysis was published only decades later (the class placed it around 2020) — it takes that long to fully document a safety disaster
- Standing lesson: drugs still reach the market without adequate safety analysis for specific subpopulations

Effectiveness and safety are two different analyses

- Every intervention is evaluated on two axes: effectiveness (the intended, favourable outcome — the trial's primary goal) and safety (adverse effects — a secondary analysis)
- Safety analysis counts adverse events and organ-level effects — and for both, a sufficiently large population is the precondition: without enough participants, rare events cannot be observed at all

Rare events and sample size: the myocarditis example, revisited

- Myocarditis after COVID-19 vaccination in young males is a very rare event
- From prior frequency estimates one can compute the lower bound of study size — the minimum number of participants needed to expect even one occurrence
- The vaccine trial was powered to detect protection from severe disease and hospitalisation, not to detect this signal; the myocarditis estimate came later, from population-scale data
- The public "betrayal" story has the same root: the vaccine's primary goal was protection from severe disease and death, not from infection or transmission; transmission prevention requires different designs and larger populations

HIV: prevalence, incidence, duration

The formula

- Prevalence = incidence × duration: prevalence accumulates everyone infected who is still alive and uncured
- For HIV the duration runs from the first registry era (late 1980s) to today, and HIV is not yet curable — so the prevalent pool only grows while treatment keeps people alive

Why rising prevalence is the success story

- About 41 million people live with HIV as of 2025 — the highest figure ever recorded
- With antiretroviral therapy (ART), life expectancy on treatment approaches that of the uninfected population: people infected in the 1990s are alive and counted
- Deaths peaked in 2004 and have fallen by roughly 73%; new infections peaked around the mid-1990s and have fallen by roughly 65%
- Deaths down, new cases down, prevalence up — the expected arithmetic of effective therapy for an incurable chronic infection, not evidence of a worsening epidemic

Transmission and the limits of the count

- Suppressed viral load on ART makes transmission unlikely — treated people largely stop transmitting (a student, Alena, made this point and it was confirmed)
- Transmission persists because not everyone has access to ART and a fraction remain undiagnosed and contagious
- The true number cannot be counted precisely: an undiagnosed fraction, unequal access, and historical undercounting under stigma — early case counts were underestimates, so historical baselines are unreliable

Vocabulary

Concept	Definition
Prevalence	All existing cases at a given time
Incidence	New cases within a defined period
Duration	How long the disease or epidemic has persisted
Prevalence = incidence × duration	The core formula, applicable to any condition
Teratogenic effect	Developmental harm to an embryo from toxins or infections
Effectiveness	The intended favourable outcome — the primary endpoint
Safety	Adverse-effect analysis — the secondary endpoint
Viral load	Concentration of virus in blood; suppressed by ART
PLHIV	People living with HIV — the term behind the 41 million

Check before you lean on it

- The notes give "10,000+ pregnant women, ~20,000 affected children"; the commonly cited figure is about 10,000 affected children worldwide. Verify against the historical review before quoting either number.
- "First registered cases: USA ~1987–88" refers to the registry era the formula used in class; the first US reports of AIDS are from 1981.